



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione



LoP (T2MP)



LoP – The Logic of Propositions

- **Introduction**
- NL to LoP + Modeling mistakes
- Truth tables

Language (the same as LoE)

Definition (Assertional language)

$$L = \langle A, FR \rangle = \{P\}$$

where:

- L is a **propositional language**, where $P \in \{P\}$ is a **proposition**.
- A = is an **alphabet** of atomic propositions
- FR is a **set of formation rules**

Formation rules – BNF

$$\begin{aligned} \langle P \rangle & ::= \langle \text{atomic proposition} \rangle \mid \\ & \quad \neg \langle P \rangle \mid \\ & \quad \langle P \rangle \wedge \langle P \rangle \mid \\ & \quad \langle P \rangle \vee \langle P \rangle \mid \\ & \quad \langle P \rangle \supset \langle P \rangle \mid \\ & \quad \langle P \rangle \equiv \langle P \rangle \mid \\ & \quad \langle P \rangle \oplus \langle P \rangle \end{aligned}$$
$$\langle \text{atomic proposition} \rangle ::= P_1 \dots P_n \in \{P\}$$

LoP Domain

Definition (Domain – facts).

$$D_{LoP} = \langle T, F \rangle$$

Definition (Domain – percepts). Let

$$D_{LoDE} = \langle E, \{C\}, \{R\} \rangle$$

be a LoDE domain of interpretation. Let $L_{LoDE} = \{a\}$ be a LoDE language for D_{LoP} where $\{a\}$ is the set of assertions in L_{LoDE} . Let $a_i \in L \subseteq L_{LoDE}$ be an assertion. Then

$$D_{LoP} = \{a^+, a^-\} = \{a_1^+, a_1^-, \dots, a_N^+, a_N^-\} = \{T(a_1), F(a_1), \dots, T(a_N), F(a_N)\}$$

where a^+, a^- are values of atomic propositions such that:

- $a^+ = T(a) = T$ if the LoDE assertion a is True
- $a^- = F(a) = T$ if the LoDE assertion a is False

Definition (Model). M is a set of atomic propositions $\{a^+, a^-\}$ such that, for each a , M contains one and only one between a^+ and a^- .

$$M = \{f\} = \{a^+, a^-\} = \{\dots, a_i^+, \dots, a_j^-, \dots\} \subseteq D_{LoP}.$$

Terminology (Model, atomic proposition). From now on, when no confusion arises, we talk of propositions meaning atomic propositions, the only propositions which belong to models.

Interpretation function

Definition (LOP Interpretation function). Let $\{P\}$ the set of atomic proposition of a LoP language. Then

$$I_{\text{LoP}}: \{P\} \rightarrow \{T, F\}.$$

Observation (Truth value of an atomic proposition). In LoP how the truth value of an atomic proposition is computed is irrelevant.

Observation (Truth value of an atomic proposition). In LoP the truth value of an atomic proposition is computed is relevant from an application point of view as it is constrained by the world as described by a LoDE EG and constrains (when a truth value is imposed on an proposition whose LoDE truth value is undefined).



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Modeling mistakes - And (1)

We express conjunctions with many words other than "**and**", including "**but**," "**moreover**," "**however**," "**although**", and "**even though**".

For example: "I enjoyed the holiday, even though it rained a lot" can be translated into the facts "I enjoyed the holiday" (E) and "It rained a lot" (R).

"I enjoyed the holiday, even though it rained a lot" $\rightarrow E \wedge R$

Sometimes "and" joins adjectives.

For example: "The leech was long and wet and slimy." This can be paraphrased as "The leech was long(L), and the leech was wet(W), and the leech was slimy(S).

"The leech was long and wet and slimy." $\rightarrow L \wedge W \wedge S$

Modeling mistakes - And (2)

Sometimes "and" does not join whole propositions into a compound proposition. Sometimes it simply joins nouns. This cannot be paraphrased. In these cases, the "and" is expressed inside the propositional variable, and not as logical connective.

For example: "Bert and Ernie are brothers". This cannot be paraphrased. "Bert is a brother and Ernie is a brother", for that does not assert that they are brothers to each other.

"Bert and Ernie are brothers" $\rightarrow B$

Modeling mistakes - Inclusive vs. Exclusive disjunction

The natural, but longwinded, way to express exclusive disjunction is

$$(\neg p \vee q) \wedge (p \vee \neg q).$$

The way to say they have different truth values is to deny their equivalence:

$$\neg (p \equiv q).$$

For example: When a menu says "cream or sugar", it uses an inclusive "or", because you may take one, the other, or both. But when it says "coffee or tea", it uses an exclusive "or", because you are not invited to take both, generally.

Modeling mistakes - Implication

$p \supset q$ translates a wide variety of English expressions, for example, "if p , then q ", "if p , q ", " p implies q ", " p entails q ", " p therefore q ", " p hence q ", " q if p ", " q provided p ", " q follows from p ", " p is the sufficient condition of q ", and " q is the necessary condition of p ". The least intuitive is " p only if q " (to be understood from $\neg q \supset \neg p$).

For example the following all translate to $p \supset q$:

- If Mario goes to the party, (then) I'll go too.
- I'll go to the party if/provided that Mario comes too.
- I'll go to the party only if Mario goes.
- Mario going to the party is the sufficient condition of me going to the party.
- Me going to the party is necessary condition of Mario going to the party.
- The decrease in white blood cells implies the antibiotic is working.

Modeling mistakes - Even If

" p even if q " means " p whether or not q " or " p regardless of q ".
Therefore one perfectly acceptable translation of it is simply " p ". If you want to spell out the claim of "regardlessness", then you could write " $p \wedge (q \vee \neg q)$ ".

For example:

- I'll go to the party even if Mario doesn't go.
- I'll go to the party whether or not Mario goes.
- I'll go to the party regardless of whether Mario comes or not

Modeling mistakes - Unless

Sometimes "unless" should be translated as inclusive disjunction ($\vee$), and sometimes as exclusive disjunction ($\oplus$).

For example (inclusive disjunction): "I'll go to the party unless I get another offer" means that I'll go if nothing else comes along, namely an exclusive disjunction. In many contexts it also means that I might go anyway; the second offer might be worse. So I'll go or I'll get another offer or both.

For example (exclusive disjunction): Consider by contrast, "I'll go to the party unless Rufus is there". In many contexts this means that if I learn Rufus is going, then I'll change my mind and not go. So either I'll go or Rufus will go but not both.

Modeling mistakes - Necessary and Sufficient conditions

We say that p is a *sufficient condition* of q when p 's truth guarantees q 's truth. By contrast, q is a *necessary condition* of p when q 's falsehood guarantees p 's falsehood.

In the ordinary material implication, $p \supset q$, the antecedent p is a *sufficient condition* of the consequent q , and the consequent q is a *necessary condition* of the antecedent p .

Notice that $p \supset q$ if and only if $\neg q \supset \neg p$.

For example: "If Socks is a cat, then Socks is a mammal". Being a cat is a sufficient condition of being a mammal. Being a mammal is a necessary condition of being a cat."

Exercise 1 - NL to LoP

Assuming A = “Angelo comes to the Party”, B = “Bruno comes to the Party”,
 C = “Carlo comes to the Party”, D = “David comes to the Party”,

David or Bruno come to the party

$$D \vee B$$

Bruno will come to the party

$$B$$

Bruno will come to the party, unless David is there

$$D \oplus B$$

Carlo comes to the party, therefore David comes too

$$C \supset D$$

Either David or Bruno come to the party

$$(D \wedge \neg B) \vee (B \wedge \neg D)$$

Neither Carlo nor David will come to the party

$$\neg C \wedge \neg D$$

Exercise 2 - NL to LoP

Assuming A = “Angelo comes to the Party”, B = “Bruno comes to the Party”,
 C = “Carlo comes to the Party”, D = “David comes to the Party”,

“If David comes to the party then Bruno and Carlo come too”

$$D \supset (B \wedge C)$$

“Angelo will come to the party, provided that Bruno comes and Carlo does not”

$$(B \wedge \neg C) \supset A$$

“Carlo comes to the party given that David doesn't come, but, if David comes, then Bruno doesn't come”

$$(\neg D \supset C) \wedge (D \supset \neg B)$$

“Carlo comes to the party only in case Angelo and Bruno do not come”

$$(\neg A \wedge \neg B) \equiv C$$

“A sufficient condition for Angelo coming to the party, is that Bruno and Carlo aren't coming”

$$(\neg B \wedge \neg C) \supset A$$

“A necessary and sufficient condition for Angelo coming to the party, is that Bruno and Carlo aren't coming”

$$(\neg B \wedge \neg C) \equiv A$$



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- **Truth Tables**

Truth table for every LoP connective

A	B	$\neg A$	$A \vee B$	$A \wedge B$	$A \supset B$	$A \oplus B$	$A \equiv B$
T	T	F	T	T	T	F	T
T	F	F	T	F	F	T	F
F	T	T	T	F	T	T	F
F	F	T	F	F	T	F	T

Satisfiability of a formula using truth tables

Is the following formula satisfiable? $\varphi: a \supset (b \supset a)$

a	b	$(b \supset a)$	$a \supset (b \supset a)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

Yes, it is actually a tautology

Satisfiability of a formula using truth tables

Is the following formula satisfiable? $\varphi: (a \vee b) \oplus (\neg b \supset (a \vee (\neg a \supset b)))$

a	b	$(a \vee b)$	$(\neg a \supset b)$	$(\neg b \supset (a \vee (\neg a \supset b)))$	$(a \vee b) \oplus (\neg b \supset (a \vee (\neg a \supset b)))$
T	T	T	T	T	F
T	F	T	T	T	F
F	T	T	T	T	F
F	F	F	F	F	F

No, the formula is unsatisfiable

Satisfiability of a formula using truth tables

Is the following formula valid? $\varphi: a \supset (b \supset \neg a)$

a	b	$(b \supset \neg a)$	$a \supset (b \supset \neg a)$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

No, the formula is not valid

Satisfiability of a formula using truth tables

Given the formulas

$$\varphi: (a \supset \neg b) \wedge \neg(c \oplus \neg a)$$

$$\psi: (c \oplus b) \supset (a \wedge b)$$

Is φ satisfiable? Yes

Is ψ satisfiable? Yes

Does $\varphi \models \psi$? No

Do φ and ψ have 2 models in common? Yes

a	b	c	$(a \supset \neg b)$	$\neg(c \oplus \neg a)$	$(c \oplus b)$	$(a \wedge b)$	φ	ψ
T	T	T	F	F	F	T	F	T
T	T	F	F	T	T	T	F	T
T	F	T	T	F	T	F	F	F
T	F	F	T	T	F	F	T	T
F	T	T	T	T	F	F	T	T
F	T	F	T	F	T	F	F	F
F	F	T	T	T	T	F	T	F
F	F	F	T	F	F	F	F	T



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